WILLIAM FORD

Email ◇ Website ◇ Scholar

ABOUT ME

I am a second-year mathematics PhD candidate in Paris at École Polytechnique and Université Paris-Saclay. My PhD is on quantitative stability of the optimal transport problem. I wrote two sole-author preprints in my first PhD year, ranked **1st of 25** in my MSc, achieved an M2 average of **18.1/20**, and ranked **7th of 129** in my BSc. Currently I am working on mathematical aspects of AI safety and alignment, as well as autoformalisation of mathematics in Lean.

EDUCATION

PhD Mathematics – École Polytechnique / Université Paris-Saclay, France 2025–2028

Research on quantitative stability of optimal transport. Supervisors: *Dr Michael Goldman and Dr Cyril Letrouit.*

Iliad Intensive – London Initiative for Safe AI, United Kingdom Nov. 2026

Four week intensive course on AI alignment. Interpretability, agent foundations, heuristics, singular learning theory, RL.

M2 Optimisation – IP Paris PhD Track / Paris-Saclay, France 2024–2025

Très Bien, 18.1/20. Stochastic optimisation, game theory, optimal transport, nonsmooth analysis, optimal control.

BSc and MSc Mathematics – Durham University, United Kingdom 2020–2024

First Class, avg. 82% and 88%. Ranked 1st of 25 on MSc. Analysis, PDEs, probability, statistics and optimisation.

RESEARCH

My PhD research is on the stability of the optimal transport problem. This is crucial to applications in machine learning and statistics, where it is important to quantify errors introduced by various approximations and estimations.

***Quantitative Stability, Coercivity and Uniqueness of Optimal Transport Plans* – preprint** Sept. 2026

In my most recent paper, I prove that when *one* of the marginal probabilities is suitably regular, the optimal transport problem is locally quantitatively stable with respect to perturbation of *both* marginals. My results also have direct implications for the statistical estimation of transport maps.

***Quantitative Uniqueness of Kantorovich Potentials* – accepted** Apr. 2026

Accepted for publication in *ESAIM: Control, Optimisation and Calculus of Variations*.

In my first paper, I gave the first quantitative bounds for almost-uniqueness of the dual optimal transport problem, as well as vastly generalising the sharp hypotheses for uniqueness to allow for lower-dimensional marginal measures. This was a result of a novel geometric insight regarding the structure of the set of optimal potentials.

***Quantitative Stability in Discrete Optimal Transport* – M2 research thesis** Oct. 2025

During my M2 research internship, I studied the stability properties of the fully discrete optimal transport problem. This has intimate connections with linear programming, and taking this novel approach to stability led to the uniqueness results in my first paper.

AWARDS, FUNDINGS, SCHOLARSHIPS

2026 Funded place on the November 2026 Iliad intensive ($5000), Iliad, Principles of Intelligence.
2024 Sophie Germain M2 Excellence Scholarship (€11,000), Fondation Mathématique Jacques Hadamard.
2024 Full M2 and PhD Track scholarship, Institut Polytechnique de Paris.
2024 Winton MSc Memorial Prize (£500), highest-scoring Mathematics MSc student, Durham University.
2021 Norton Prize, performance in science, St Cuthbert's Society, Durham University.

QUANTITATIVE AND TECHNICAL SKILLS

Mathematics Probability, stochastic analysis, optimisation, statistics, optimal control, partial differential equations, information theory, ergodic theory, mathematical finance, linear algebra, calculus, numerical analysis.
Programming Python (NumPy, PyTorch, Matplotlib), Lean, AI-assisted workflows, LaTeX.
Spoken languages English (native), French (fluent), German (intermediate), Italian (learning).
Interests Road and mountain biking, swimming, classical & jazz trumpet, Cuban salsa, close-up magic.

TEACHING

Teaching Assistant – École Polytechnique 2025–2026

Teaching Assistant – Durham University 2023–2024

Mathematics Tutor – MyTutor.co.uk 2021